Week 5 Report

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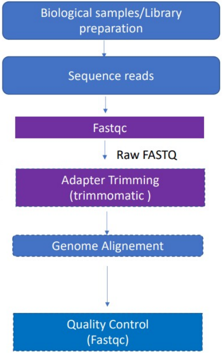
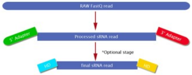
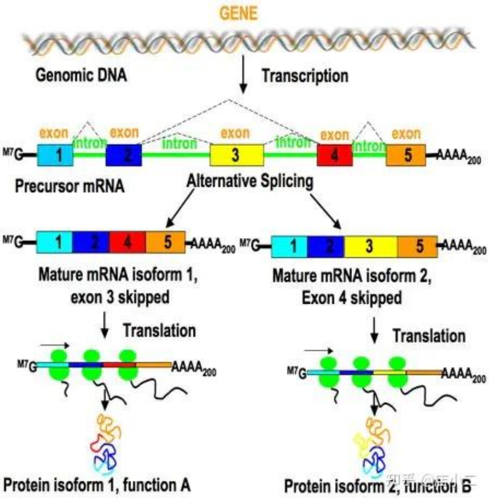
Read


Biological samples/Library preparation
Sequence reads
Fastqc
Raw FASTQ
Adapter Trimming (trimmomatic)
Genome Alignement
Quality Control (Fastqc)

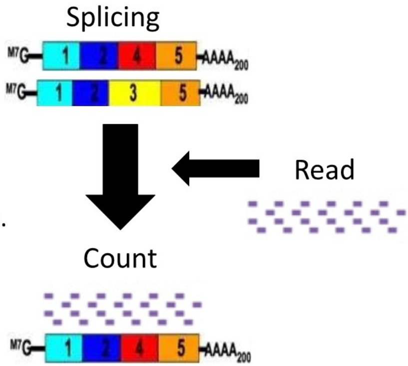

RAW FastQ read
5' Adapter
Processed sRNA read
3' Adapter
*Optional stage
HD
final sRNA read
HD

Alternative Splicing

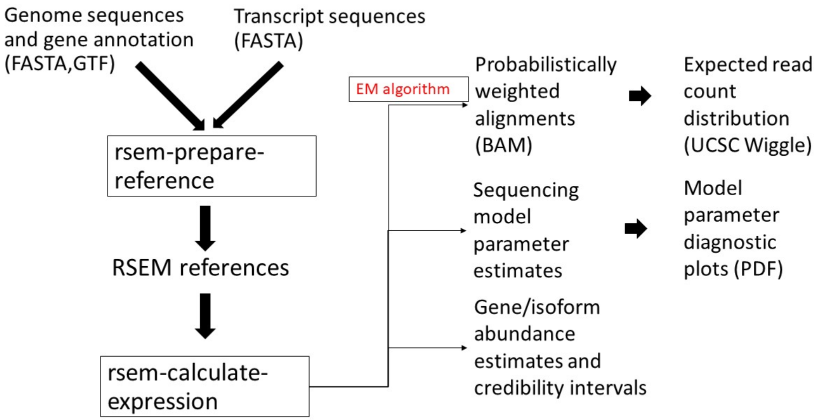


count

- Throws away data and produces biased estimates if "mappability" is not taken into account.
- Produces incorrect estimates for alternatively-spliced genes.
- Does not extend well to the task of estimating isoform abundances.



RSEM


Genome sequences and gene annotation (FASTA,GTF)
Transcript sequences (FASTA)
rsem-prepare-reference
RSEM references
rsem-calculate-expression
EM algorithm
Probabilistically weighted alignments (BAM)
Expected read count distribution (UCSC Wiggle)
Sequencing model parameter estimates
Model parameter diagnostic plots (PDF)
Gene/isoform abundance estimates and credibility intervals

1. RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome

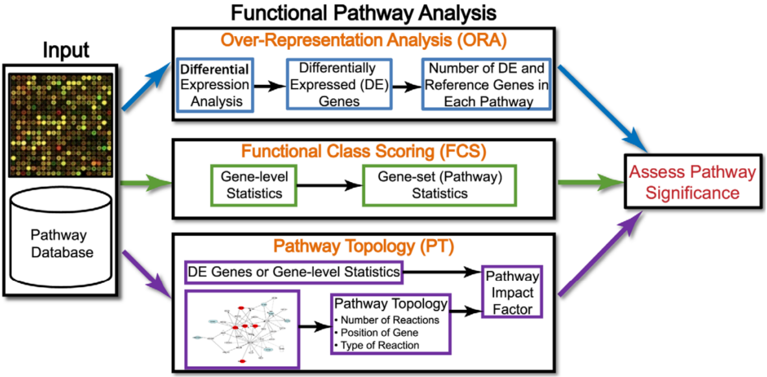
2. Pathway analysis

Pathway Analysis Methods

Goal :

1. Looking at DEGs alone often can' t explain the biological mechanism. Pathway analysis, using databases like KEGG or Reactome, helps identify which pathways and functions these genes are involved in.
2. It detects which pathways (e.g., cell cycle, apoptosis, immune signaling) are significantly changed between different conditions, treatments, or diseases.
3. It helps summarize which cellular behaviors or biological processes may be affected based on the DEGs.

Pathway Analysis Methods



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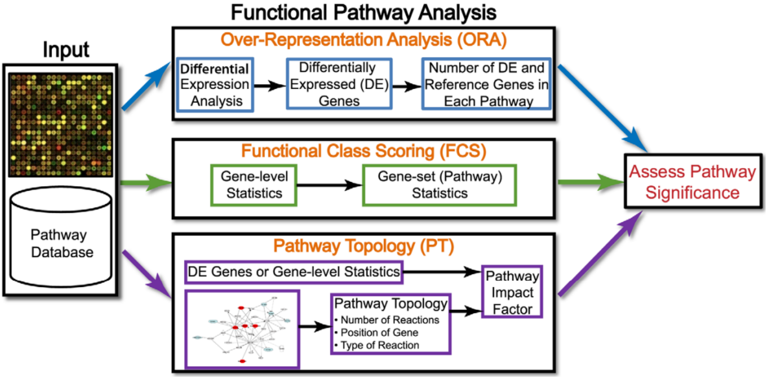
[1] Purvesh Khatri, Marina Sirota, and Atul J Butte. "Ten years of pathway analysis: current approaches and outstanding challenges". In: *PLoS computational biology* 8.2 (2012), e1002375.

Method Limitations

ORA Limitations :

1. Ignore any values associated with them such as probe intensities.
2. Using an arbitrary threshold lead information loss.
3. Assuming independence between genes amounts to **competitive null hypothesis testing** (see below), which ignores the correlation structure between genes.
4. Assumes that each pathway is independent of other pathways, which is erroneous.

Pathway Analysis Methods

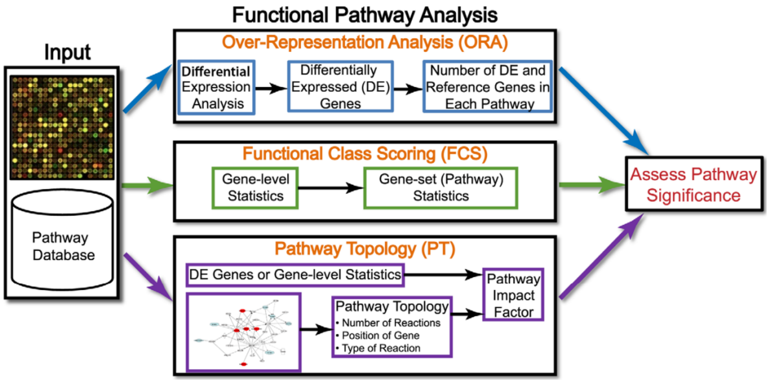


Method Limitations

FCS Limitations :

1. Assumes that each pathway is independent of other pathways, which is erroneous.
2. Rank-Based or Statistic-Based

Pathway Analysis Methods



Pathway Topology

1. ScorePAGE [2] :
   Computes the similarity between each pair of genes in a pathway. This similarity is analogous to the gene-level statistics used in FCS methods.
2. Impact Factor (IF) analysis [3] :
   IF analysis models a signaling pathway as a graph, defines a gene-level statistic, called perturbation factor (PF) of a gene.

$$PF(g_i) = \Delta F(g_i) + \sum_{j=1}^{n} \beta_{ji} \cdot \frac{PF(g_j)}{N_{ds}(g_j)}$$

   Then Calculate IF :

$$IF(P_i) = log\left(\frac{1}{p_i}\right) + \frac{|\sum_{g \in P_i} PF(g)|}{N_{de}(P_i)}$$

Pathways are not static, but change their interaction structure according to physiological conditions.

NetGSA [4] accounts for the the change in correlation as well as the change in network structure as experimental conditions change.

1. accounts for a gene’s baseline expression.
2. the pathways be represented as directed acyclic graphs.

Pathway Analysis

Category :

1. Knowledge-based approaches
2. Data-driven approaches
3. Multi-omics pathway analysis
4. Dynamic / temporal pathway analysis
5. Single cell pathway analysis

參考文獻 I

[1] Purvesh Khatri, Marina Sirota, and Atul J Butte. "Ten years of pathway analysis: current approaches and outstanding challenges". In: *PLoS computational biology* 8.2 (2012), e1002375.

[2] Jörg Rahnenführer et al. "Calculating the statistical significance of changes in pathway activity from gene expression data.". In: *Statistical Applications in Genetics & Molecular Biology* 3.1 (2004).

[3] Sorin Draghici et al. "A systems biology approach for pathway level analysis". In: *Genome research* 17.10 (2007), pp. 1537–1545.

參考文獻 II

[4] Ali Shojaie and George Michailidis. "Analysis of gene sets based on the underlying regulatory network". In: *Journal of Computational Biology* 16.3 (2009), pp. 407–426.